Content Delivery Network (CDN)

A CDN is a geographically distributed network of servers that delivers web content from locations close to users

- **Cache hit:** content is served directly from an edge server, reducing latency and origin load

- **Cache miss:** the edge server retrieves the content from the origin, potentially caching it for future requests

➡ Typically static assets (HTML, CSS, JavaScript, images, videos), but also cacheable API responses

Example: *Akamai*, *Cloudflare*

Pors and Cons of CDN

Benefits

- ✓ Lower latency for geographically distributed users
- ✓ Reduced bandwidth and computational load on origin servers
- ✓ Improved scalability and availability; many CDNs also provide DDoS protection

Limitations

- ◉ Cache invalidation and stale content; personalized responses must not be inadvertently shared between users
- ◉ Cost